

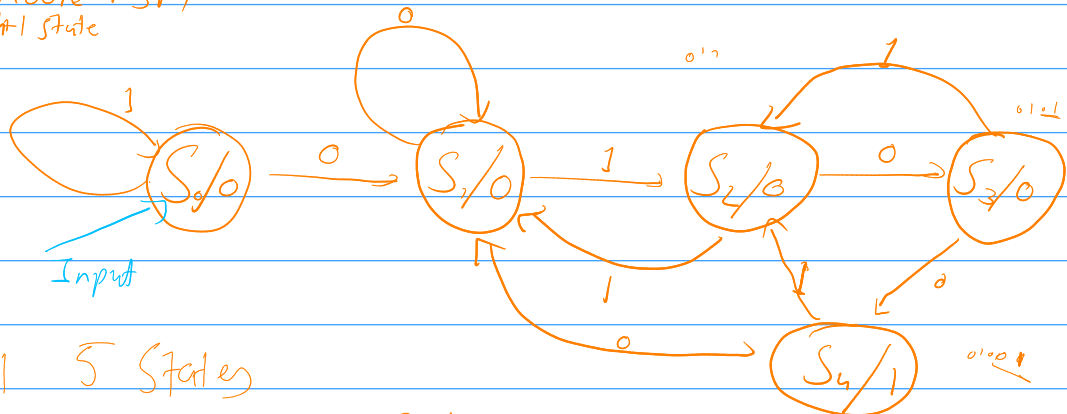
Digital Assignment III

1 Moore and Mealy FSM for sequence detector ^{MSB} 0010 ^{LSB}. First received bit is LSB

Ans

4th 3rd 2nd 1st
0 → 0 → 1 → 0

Moore FSM
4+1 state



Total 5 States

S₀ 000
S₁ 001
S₂ 010
S₃ 011
S₄ 100

State table

PS	NS		
	X=0	X=1	X=0
S ₀	S ₁	S ₀	0
S ₁	S ₁	S ₂	0
S ₂	S ₃	S ₁	0
S ₃	S ₄	S ₂	1
S ₄	S ₁	S ₂	0

So now we have 4 states

Q_A Q_B
S₀ 0 0
S₁ 0 1
S₂ 1 0
S₃ 1 1

S ₀	S ₁	S ₀	0
S ₁	S ₁	S ₂	0
S ₂	S ₃	S ₁	0
S ₃	S ₁	S ₂	1

Excitation table

Q _A	Q _B	X	Q _A ⁺	Q _B ⁺	Y	D _A	D _B
0	0	0	0	1	0	0	1
0	0	1	0	0	0	0	0
0	1	0	0	1	0	0	1
0	1	1	1	0	0	1	0
1	0	0	1	1	0	1	1
1	0	1	0	1	0	0	1
1	1	0	0	1	1	0	1
1	1	1	1	0	0	1	0

$$D_A = \overline{A}BC + A\overline{B}C$$

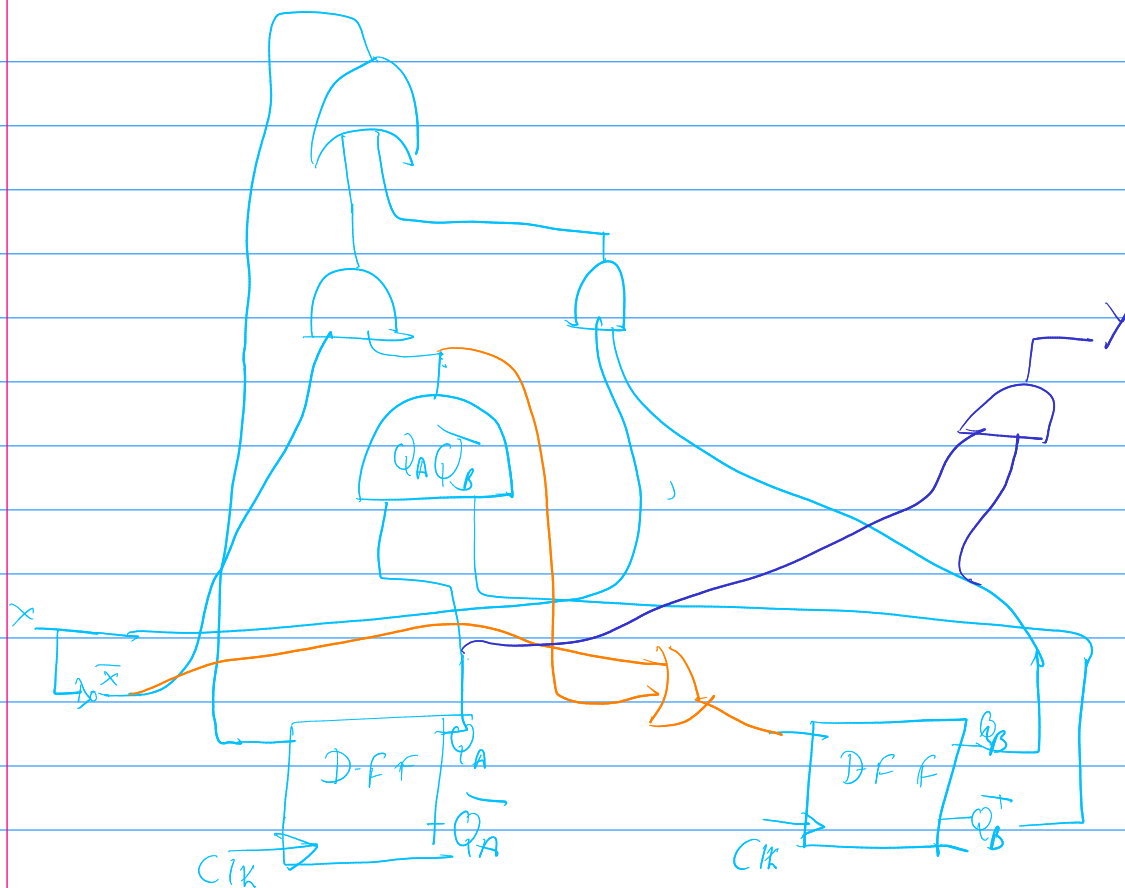
	00	01	11	10
0	0	0	1	0
1	1	0	1	0

$$D_A = A\overline{B}C' \approx Q_A Q_B' X' + Q_B X$$

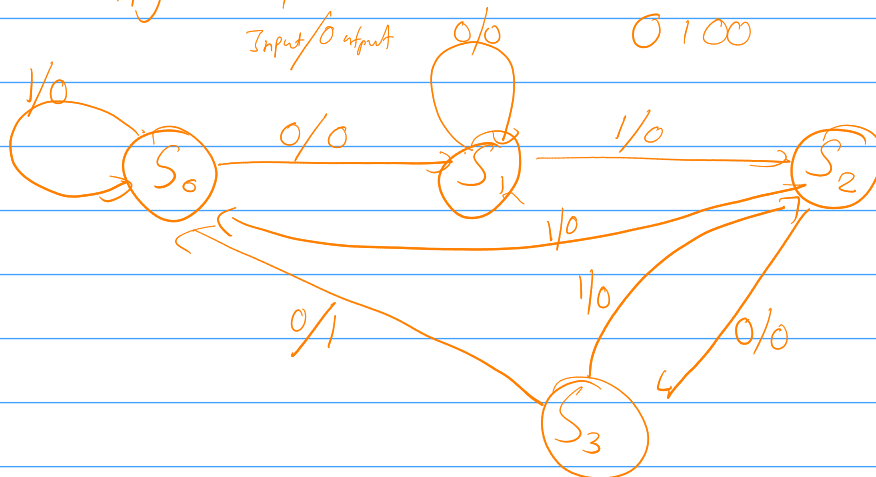
$$D_B = \overline{Q_A} \overline{Q_B} X + Q_A Q_B$$

	00	01	11	10
0	1	0	0	1
1	1	0	0	1

$$Y = Q_A Q_B$$



Mealy FSM 4 state



Mealy Excitation table

PS	X	NS	Y
S ₀	0	S ₁	0
S ₀	1	S ₀	0
S ₁	0	S ₁	0
S ₁	1	S ₂	0
S ₂	0	S ₃	0
S ₂	1	S ₀	0
S ₃	0	S ₀	1
S ₃	1	S ₂	0

NS

PS	X=0	X=1	Y	X=0	X=1
S ₀	S ₁	S ₀	0	0	0
S ₁	S ₁	S ₂	0	0	0
S ₂	S ₃	S ₀	0	0	0
S ₃	S ₀	S ₂	1	0	0

No further reduction

4 State = 2 FF

S ₀	00
S ₁	01
S ₂	10
S ₃	11

Q _A	Q _B	X	Q _A ⁺	Q _B ⁺	D _A	D _B	Y
0	0	0	0	1	0	1	0
0	0	1	0	0	0	0	0
0	1	0	0	1	0	1	0
0	1	1	1	0	1	0	0
1	0	0	1	1	1	1	0
1	0	1	0	0	0	0	0
1	1	0	0	0	0	0	1
1	1	1	1	0	1	0	0

$$Y = Q_A Q_B$$

D_A

A	B	X	D _A
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

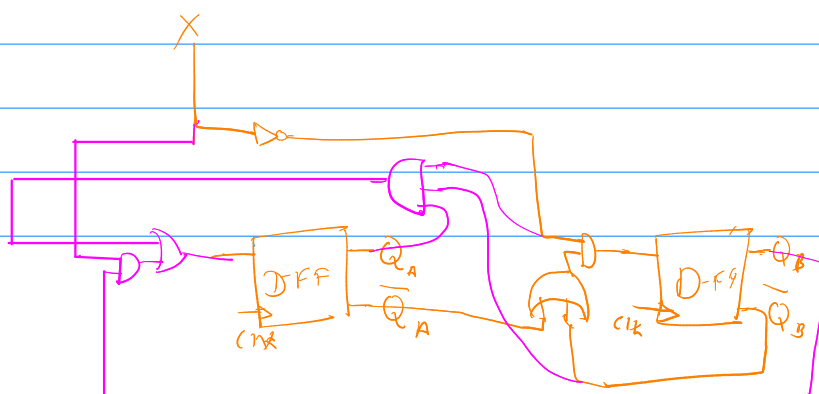
$$= Q_A \bar{Q}_B \bar{X} + Q_B X$$

D_B

A	B	X	D _B
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

$$= \bar{Q}_A \bar{X} + \bar{Q}_B \bar{X}$$

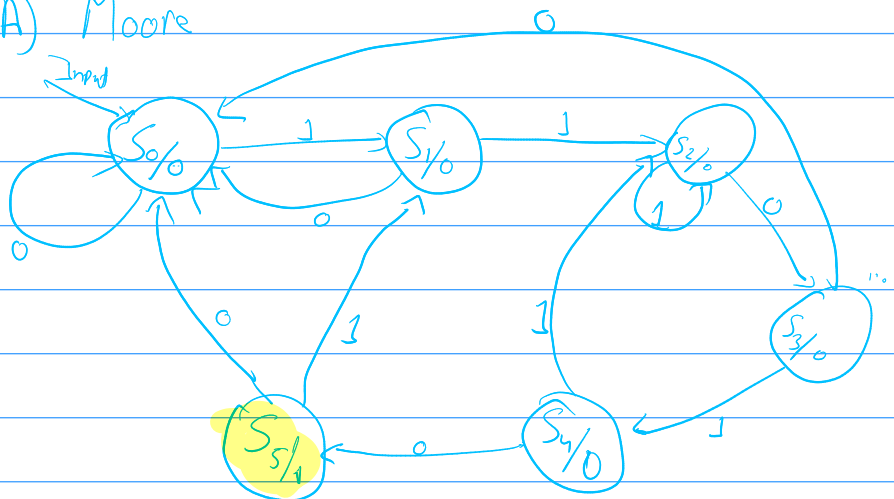
$$= \bar{X} (\bar{Q}_A + \bar{Q}_B)$$



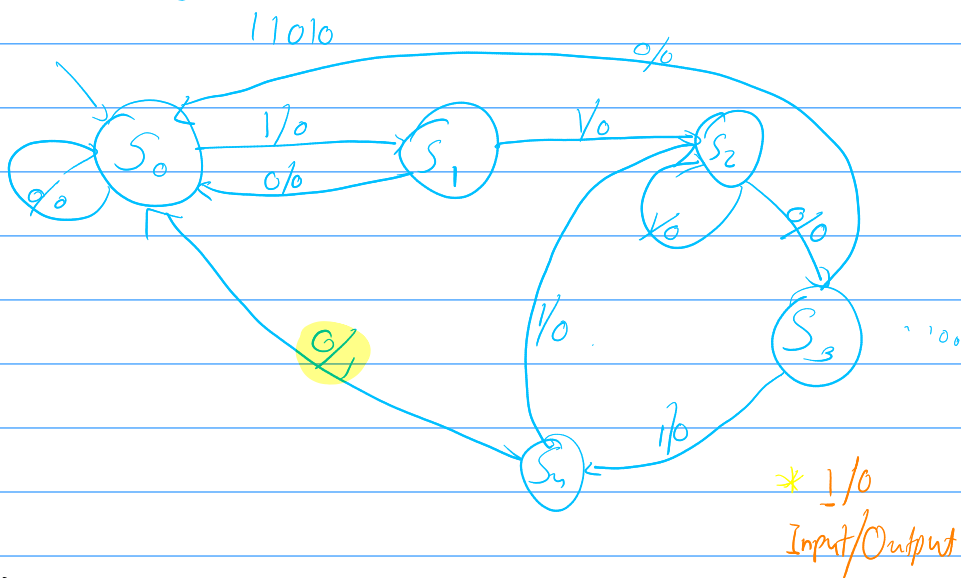
Q.2 Moore & Mealy State diagram for below sequence

i) 11010

A) Moore

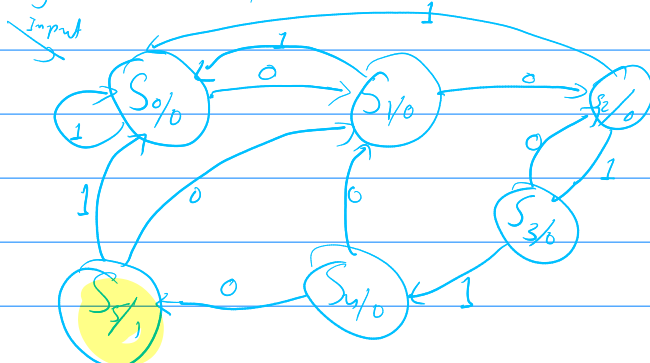


B) Mealy

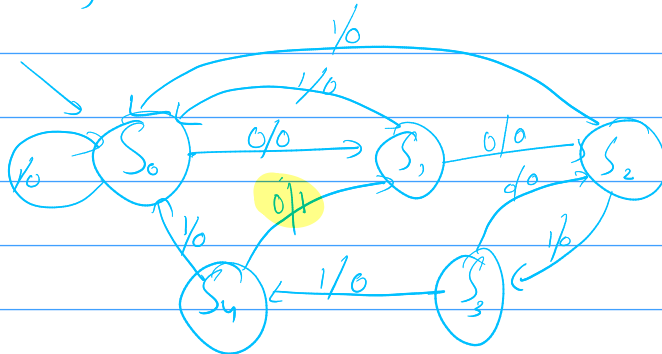


2) 00110

A) Moore FSM

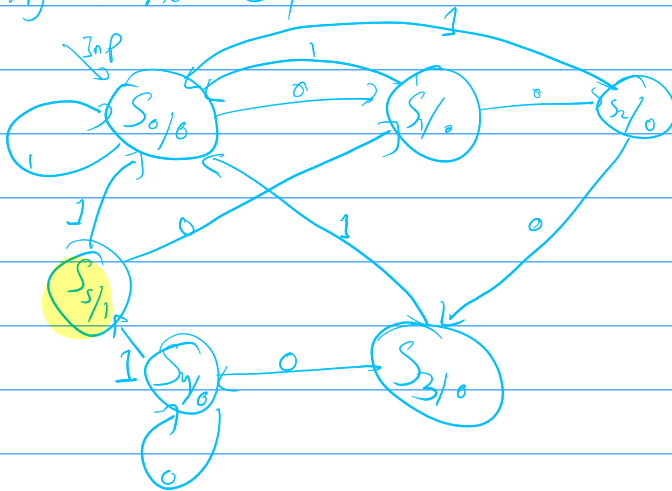


B) Mealy FSM 00110

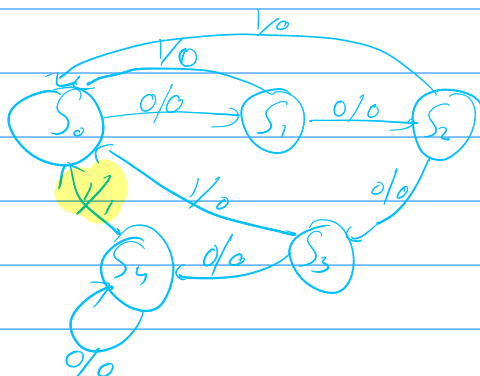


3) 00001

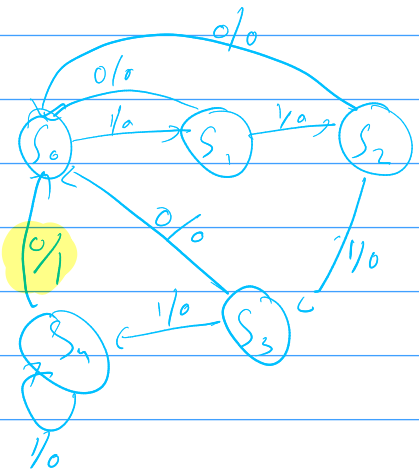
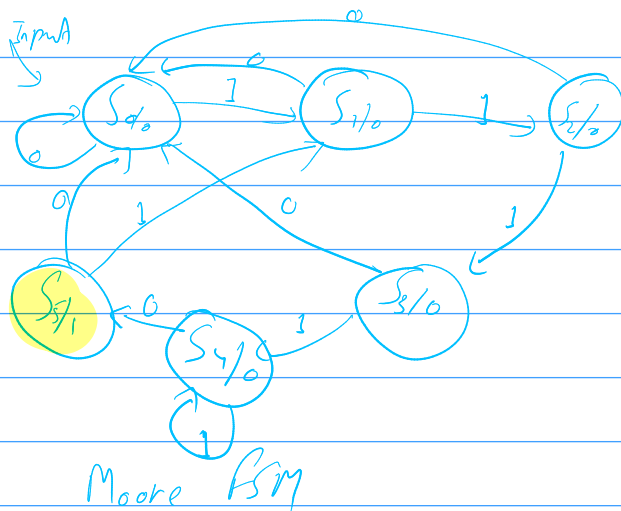
A) Moore FSM



B) Mealy FSM 00001

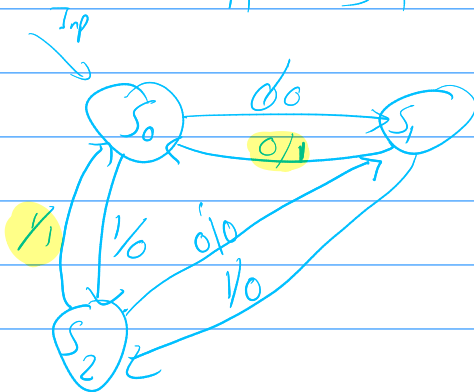


4) 11110

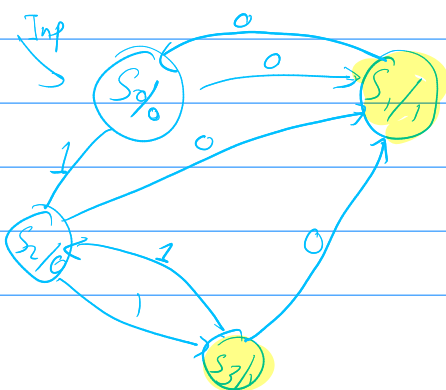


Q3 FSM have Input 1 Output, Output = 1 when input has pair of 0's or pair of 1's

00 → 1
11 → 1

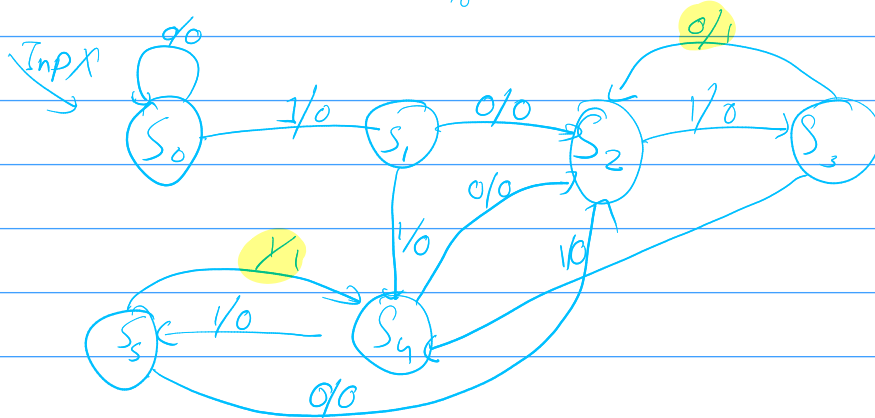


Q4 Input W Output Z
Z=1 if odd 0's or even 1's Draw Moore FSM



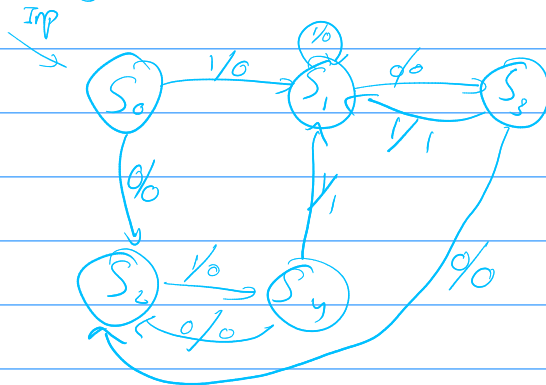
Q5 Input x Output F
 $F=1$ if $x=1010, 111$

Mealy FSM



Q6 Detect 101 or 011

Mealy FSM

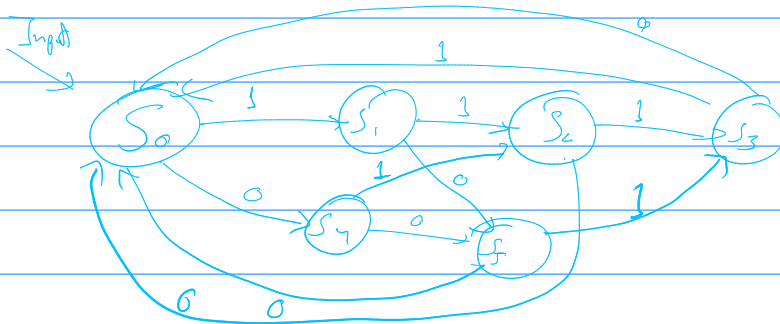


1 1 1
 101101101
 1 1
 10000

Q7 Modulo-3 counter when $w=0$

Modulo-4 counter when $w=1$

w Input

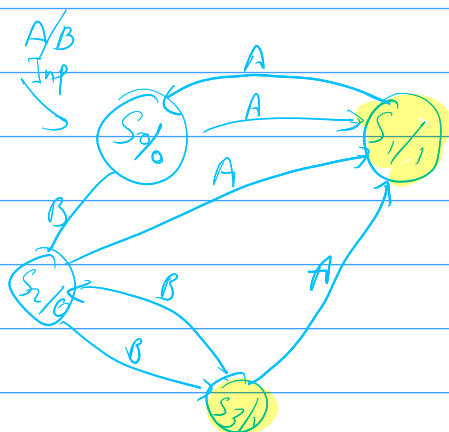


Q 8 Output Z

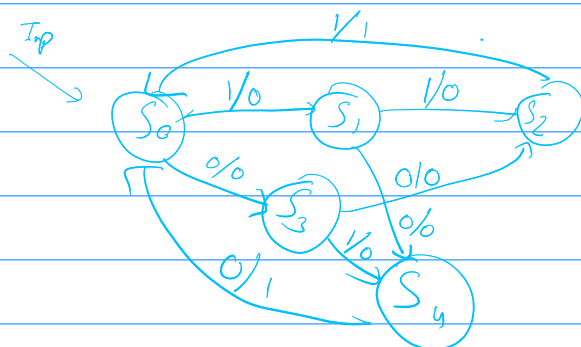
$$Z = 1$$

Even B's or Odd A's

A, B can be input



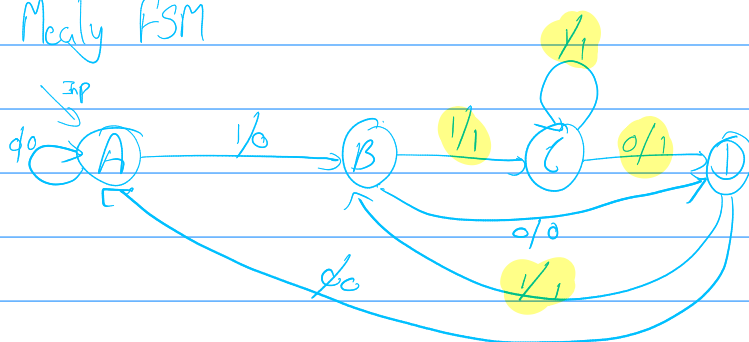
Q-9 3 bit Parity Generator



0
 ✓ 1 1 1
 ✓ 1 0 0
 ✓ 0 1 0
 ✓ 0 0 1

Q-11 fsm to detect more than one 1's in last 3 sample

Mealy FSM



0 1 0 1 0 1 1 0 0
 0 0 0 1 0 1 1 0 0

Possible Solution { 0 1 1 0 1 1 0 0 }

Q-10 Design circuit generate sequence 0, 1, 1, 1, 0, 0, 1, 0, 1